Robust Consultation for Option-Preserving Assistance

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Abstract

Acting immediately can consume an option that a beneficiary would have preferred to keep. In a two-stage decision model, preserving a menu and learning the beneficiary's state yields a value of information with an explicit cost threshold. When both the beneficiary's likely preference and the reliability of a reply are uncertain, we derive the maximin rule for rectangular uncertainty. When the two uncertainties are coupled, mixing immediate assistance with consultation can strictly improve the guarantee. A finite exact-arithmetic audit checks both settings. For any finite joint scenario set, a two-dimensional distance formula gives the exact optimal guarantee. For a finite batch, we distinguish a hard question quota from its average-rate counterpart. When the scenario can respond to the remaining quota, a backward recursion prices the loss of one question slot.

Summary and scope

Question. When is it worth asking a beneficiary which already permitted option to use?

Finding. The value depends on the joint uncertainty about preferences and reply quality. Its convex hull gives a sharp guarantee, and randomizing between asking and acting can improve that guarantee.

Evidence and scope. Exact checks cover rectangular and coupled models. The fitted comparison uses synthetic costs and labels. A reply can inform the choice within the authorized menu, and permission is checked separately.

1 Why keeping an option changes the decision

An assistant can offer two possible ways to resolve a problem, prepare either one, or commit to one immediately. Suppose permitted actions form the same finite menu $\mathcal{A}$ in each state θ. Acting immediately yields

$$V_0 = \max_{a \in \mathcal{A}} E[u(a, \theta)].$$

If a reversible step preserves the menu and a query perfectly reveals θ before action, the expected value after paying query cost c is $V_1 = E[\max_a u(a, \theta)] - c$. This is a value-of-information comparison in the tradition of Blackwell's order [1], with an explicit cost for the more informative experiment. An irreversible operation can remove later options, so the calculation assumes a preserved menu.

Proposition 1. *The reversible and informed route strictly improves expected utility exactly when* $c < E[\max_a u(a, \theta)] - \max_a E[u(a, \theta)]$. *The right side is nonnegative.*

Proof. Subtract V_0 from V_1. For every fixed action the statewise maximum is at least its utility, so expectation dominates the best fixed action. Equality holds whenever the same action maximizes in every positive-probability state. □

2 An exact threshold

Take two equally likely states. A tailored choice yields 4 and the wrong choice yields -2. An immediate commitment gives 1, while perfect information yields 4 before cost. The threshold is 3. At cost 2 the reversible route is better, and at cost 4 acting immediately has higher expected utility within this toy contract. With a symmetric answer channel of accuracy $\rho \geq 1/2$, using the indicated option yields $6\rho - 2$. Its improvement before cost is $6\rho - 3$. A bare instruction to always ask would lose in several of these cases.

3 When consultation reliability is uncertain

Suppose both actions are permitted and exactly one matches the beneficiary's state. Normalize the matching and mismatching payoffs to one and zero, respectively, and write $d \geq 0$ for consultation cost in those units. Let p be the probability that option one matches. The binary answer is correct with the same probability r in either state. A scenario (p, r) is fixed before the policy draws any private random coin. The policy knows the uncertainty set, and the reply is its only observation after choosing to ask.

The four relevant deterministic payoffs are

$$p, \qquad 1 - p, \qquad r - d, \qquad 1 - r - d.$$

They correspond to choosing option one immediately, choosing option zero immediately, following the reply, and reversing the reply. A policy that pays to ask and then always chooses the same option is weakly dominated by its immediate counterpart.

Proposition 2. *For an uncertainty rectangle $[\ell, u] \times [r_-, r_+]$, the optimal worst-case expected payoff is*

$$\max\bigl\{\ell, 1 - u, \tfrac{1}{2}, r_- - d, 1 - r_+ - d\bigr\}.$$

Proof. The first two values are attained by the immediate options. An equal mixture of those options attains $1/2$ in every scenario. Following or reversing the reply attains the last two values, respectively. For an upper bound choose the admissible scenario whose p and r are each closest to $1/2$. At that scenario the best immediate payoff is $\max\{\ell, 1 - u, 1/2\}$ and the best query payoff is $\max\{r_- - d, 1 - r_+ - d, 1/2 - d\}$. Averaging among policies cannot exceed the highest payoff there. Since $d \geq 0$, the claimed formula follows. □

The rectangle can be too pessimistic if the weak regimes for immediate action and consultation cannot occur together. For example, at $d = 1/10$ take the two scenarios $(p, r) = (19/20, 11/20)$ and $(11/20, 19/20)$. The best pure policy guarantees $11/20$. An equal mixture of immediate option one and following the reply gives $7/10$ in both scenarios. To certify optimality, give the two scenarios weights $3/8$ and $5/8$. Each of those two policies then has mean $7/10$, and the other two have smaller means. Every policy's worst-case payoff is bounded above by its mean under these weights. The enclosing rectangle gives only $11/20$, as it permits low values of both p and r at once.

4 The exact value of a joint scenario set

Let S be any finite nonempty set of pairs (p, r) under the same symmetric answer and cost assumptions. For a fixed policy, its expected payoff is affine in these two coordinates. Consequently the worst payoff over S agrees with the worst payoff over its convex hull.

Proposition 3. *The optimal worst-case expected payoff is*

$$\frac{1}{2} + \min_{(p,r)\in \operatorname{conv} S} \max\left\{ \left| p - \frac{1}{2} \right|, \left| r - \frac{1}{2} \right| - d,\ 0 \right\}.$$

Proof. The finite game has four pure policies and the scenarios in S as its opposing choices. A randomized policy is a probability vector over the four policies. Linear-programming duality equates its best worst-case payoff to the smallest best-policy payoff under a mixture of scenarios. The mixture is represented by its mean pair $(p, r) \in \operatorname{conv} S$ because every pure payoff is affine. The maximum of the four pure payoffs at that pair is $\max\{p, 1-p, r-d, 1-r-d\}$. Subtracting $1/2$ gives the displayed distance expression. □

Geometrically, expand the rectangle $|p - 1/2| \le \gamma$, $|r - 1/2| \le d + \gamma$ from $\gamma = 0$ until it meets the hull. The first intersection gives the guaranteed advantage γ over a fair immediate choice. Inspecting individual scenarios can miss the least favorable mixture in the interior. For example, the triangle with vertices $(3/4, 1/2), (1/4, 3/4), (1/4, 1/4)$ contains $(1/2, 1/2)$ in its interior. At zero query cost the robust value is $1/2$, although checking only its three edges gives a larger value. On 96 three- and four-scenario fixtures the hull calculation agreed with independent enumeration of the policy linear program's vertices. Mixed policies strictly improved on pure policies in 78 fixtures. The counts concern the declared synthetic grid.

In a consent-based workflow, the alternatives could be an immediately usable, reversible draft and a draft tailored after a short question. Joint measurements of preference prevalence and reply reliability can support a scenario set. Use the rectangle when the evidence supports only marginal ranges. A stronger guarantee for a smaller joint set depends on that set covering plausible cases. These policies neither authorize new actions nor guarantee the welfare of each beneficiary.

5 A limit on how often to ask

A service may want a guarantee without interrupting every person. Let $q \in [0, 1]$ bound the probability of asking. This is distinct from d, the utility cost each time the service asks. On the scenario hull write $A(p) = \max(p, 1-p)$ for the best immediate value and $B(r) = \max(r, 1-r) - d$ for the best query value.

Proposition 4. *With a question rate at most q, the optimal worst-case payoff is*

$$V(q) = \min_{(p,r)\in \operatorname{conv} S} \big[A(p) + q \max(0, B(r) - A(p)) \big].$$

A target $v > V(1)$ is infeasible. For $v \le V(1)$ the smallest permitted rate is

$$\sup_{(p,r)\in \operatorname{conv} S:\ A(p) < v} \frac{v - A(p)}{B(r) - A(p)},$$

with value zero if the set is empty.

Proof. At a fixed scenario mixture, let $x \le q$ be total probability assigned to query policies. The best payoff at that x is $(1-x)A(p) + xB(r)$, since the two kinds of actions can be chosen separately. Its maximum over x is the stated expression. The capped policy set is a compact polytope. The same finite-game duality as before interchanges maximization over its vertices and minimization over the scenario hull. If $A(p) \ge v$, the point imposes no lower bound on q. Otherwise, feasibility at $q = 1$ implies $B(r) \ge v > A(p)$, and the expression is at least v exactly when $q \ge (v - A(p))/(B(r) - A(p))$. Every point must satisfy its inequality, which yields the supremum. □

The expression in the minimization is the maximum of six affine functions. Two represent immediate choices. Four represent asking with probability q and choosing one immediate action and one reply rule. Thus the computation can use the vertices where active affine functions change across the hull. A separate policy oracle instead enumerates mixtures of the original four choices subject to the explicit rate constraint.

In the earlier pair of scenarios with query cost $1/10$, a one-quarter rate gives $5/8$ instead of the unrestricted $7/10$. To guarantee $13/20$ requires a rate of one third. A rate above one half gives no additional gain in this case. Across 256 exact comparisons, the capped hull result agreed with the policy oracle. A one-quarter cap strictly reduced the unrestricted value on 14 of 64 scenario sets.

6 A hard quota for a finite batch

Suppose n cases share a quota of $k \le n$ questions. Each case has the same authorized menu, cost and scenario set S. The objective is the sum of case utilities. A scenario is fixed before the policy draws its current coins. An oblivious scenario sequence is fixed before the batch. A predictable adversary may instead choose each scenario after observing the remaining quota. It has no access to the policy's future coins. Neither party has a persistent hidden state that constrains the future scenario set.

Proposition 5. *For oblivious scenarios the optimal per-case guarantee under the hard quota is $V(k/n)$. For predictable scenarios let $G(n,k)$ be the optimal total guarantee, with $G(0,k) = 0$, $G(n,0) = nV(0)$, and $G(n,k) = G(n,n)$ for $k > n$. Define*

$$U(c) = \min_{(p,r)\in \operatorname{conv} S} \max\{A(p), \max(r, 1-r) - c\}.$$

For $n, k \ge 1$,

$$G(n,k) = G(n-1,k) + U\big(d + G(n-1,k) - G(n-1,k-1)\big).$$

Proof. For a fixed sequence, take a rate-capped optimal policy with question probability $x \le k/n$. Pick k distinct slots uniformly. For $k > 0$, ask independently in each eligible slot with probability nx/k. The conditional action mixtures match the optimal one-case policy and each slot has the intended marginal. This delivers $V(k/n)$ while respecting the quota on every path. With $k = 0$, always use the immediate mixture. For the converse, draw the scenarios independently from a least favorable mixture attaining $V(k/n)$. Past observations cannot predict a fresh scenario draw. If the expected fraction of questions is $y \le k/n$, the per-case payoff under the mixture is at most $(1-y)A + yB$. Its maximum over $y \le k/n$ equals $V(k/n)$. Some fixed sequence therefore gives at most this value.

In the predictable game, retaining a slot leaves continuation $G(n-1,k)$. Spending it leaves $G(n-1,k-1)$. The difference is nonnegative by inclusion of feasible policies. Conditional on the public history, all future scenarios remain possible. Subtracting the common continuation therefore raises the effective question cost by that difference. The one-case minimax identity gives U at the raised cost. Backward induction supplies strategies for both players at every quota. □

For the earlier two-scenario example, put $Z_n \sim \mathrm{Bin}(n, 1/2)$. The predictable value has the exact form

$$G(n,k) = \frac{11n}{20} + \frac{3}{10} E[\min(k, Z_n)].$$

To see this, the expression's marginal value of a slot is $(3/10)P(Z_n \ge k)$. For any such continuation charge Δ between zero and $3/10$, immediate option one and following the reply dominate their

counterparts. Their worst-case mixture is optimized by asking with probability $1/2$ when a slot remains. The two scenario payoffs meet there and the stage value is $7/10 - \Delta/2$. The recurrence for $E[\min(k, Z_n)]$ under one additional fair coin yields the displayed value and the stated boundary cases.

With two cases and one question, the oblivious guarantee is $7/10$ per case. The predictable guarantee is $53/80$. At $k = n/2$ for even n, the total gap is $(3/20)E|Z_n - n/2|$, so the gap per case is at most $3/(40\sqrt{n})$. This follows from symmetry and the binomial variance. The verifier compared 336 reachable states using two exact stage solvers, checked the binomial expression at 560 states, and checked the two-case saddle against all eight scenario trees and twelve pure plans.

A support team with a fixed daily question allowance can randomly reserve slots if its case mix is fixed before those choices. If requests arrive in response to disclosed capacity, use a state-dependent guarantee. Neither calculation changes the permission needed to act within an individual case.

7 Building examples that transfer

Generate episodes with identical first messages and different beneficiary preferences. Include a reversible preparatory step, an immediate commitment, and a short question with an explicit delay or attention cost. In independent test families measure whether the model retains meaningful control for the affected person and whether it delivers helpful work when a question has little value. Permission is a separate constraint on the menu, including any cost of delay in time-sensitive cases. Collect preferences under consent and keep emergency scenarios subject to appropriate authority.

8 Verification and limits

The checker compares the statewise formula against explicit enumeration of four contingent policies across all two-by-two tables over $\{-2, 0, 4\}$ and five priors. All 405 agree. An exact second audit checked the robust value on 270 rectangles and compared primal policy mixtures with dual scenario bounds on 315 scenario-pair and cost combinations. Mixing strictly improved on the best pure policy in 118 cases. These counts describe the fixed rational grid. The proposal depends on realistic costs and faithful preferences. A falsely reversible operation, an answer channel with state-dependent error, or an omitted stakeholder can invalidate the model.

9 Training and testing when to ask

The learned logistic policy trained on 320 synthetic examples per seed using the rule $6\rho - 3 > c$. It chose whether to pay for information. Across 3,200 fresh episodes, it queried 1,690 times. It achieved mean realized utility 1.512, compared with 1.094 for always asking and 0.976 for never asking. All policies shared the same realized consultation and immediate-action outcomes for each episode. The per-seed improvement over never asking ranged from 0.199 to 0.857 utility units. The evaluation depends on an available, reversible menu and the simulated accuracy and cost distribution. It supplies an empirical check on the decision rule in that environment.

Review status. Independent manuscript review and a language-model study remain open.

References

[1] Johannes Rauh, Pradeep Kr. Banerjee, Eckehard Olbrich, Jürgen Jost, Nils Bertschinger, and David H. Wolpert. Coarse-Graining and the Blackwell Order. *Entropy* 19(10), 527, 2017. `https://arxiv.org/pdf/1701.07602`.